XRE supporting partitions in C++ for XNG

FENTISS | REBECCA | 4.3 | Version 1.5

Schema Version		
		0.9
Basic Information		
IP Name		XRE supporting partitions in C++ for XNG
Version		1.5
Project		REBECCA
Work Item		4.3
TRL		TRL-4
Status		Demonstrated
License		Proprietary
IP Type	Category	• Software • Hypervisor
IP Type	Domain	• Safety • Security
IP Type	Comment	XRE has been evolved to support C++-based applications in a partition over XNG. This will allow C++-based applications to be executed as a bare metal application over the XNG software stack, leveraging the safety features that the hypervisor provides to the system
Repository URL		www.fentiss.com
Provider	Item 1 / Organization Name	Fent Innovative Software Solutions S.L.
Provider	Item 1 / Organization URL	www.fentiss.com
Provider	Item 1 / Organization Alias	• FENTISS
System-Level Information		
Architecture		RISC-V
Processor		CVA6
ISAExtension		rv64imafdc_zicsr_zifencei_sv39
Board		U55C Alveo FPGA
SoC		NA
SW Dependencies		None
Requires Specialized Tool		No
Interfaces and Artifacts		
API		• Standard XNG hypercalls
Input Artifacts	Item 1 / Artifact	Hypervisor configuration
Input Artifacts	Item 1 / Format	XML
Input Artifacts	Item 2 / Artifact	Partition(s) source code

Input Artifacts	Item 2 / Format	C++
Output Artifacts	Item 1 / Artifact	System image
Output Artifacts	Item 1 / Format	ELF
Deployment		
Bundle		Operational software package